

my_first_book

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Preface

This is a Quarto book.

To learn more about Quarto books visit <https://quarto.org/docs/books>.

```
1 + 1
```

```
[1] 2
```

1 Introduction

This chapter demonstrates dynamic R computation, data exploration, and cross-referencing.

1.1 Exploring Motor Trend Data

We analyze the classic `mtcars` dataset. First, we inspect a styled summary table shown in Table 1.1.

```
knitr::kable(head(mtcars[, 1:4], 5))
```

Table 1.1: Top 5 rows of Motor Trend Car Road Tests

	mpg	cyl	disp	hp
Mazda RX4	21.0	6	160	110
Mazda RX4 Wag	21.0	6	160	110
Datsun 710	22.8	4	108	93
Hornet 4 Drive	21.4	6	258	110
Hornet Sportabout	18.7	8	360	175

Next, we inspect the relationship between engine displacement and fuel efficiency in Figure 1.1.

```
plot(  
  mtcars$disp, mtcars$mpg,  
  xlab = "Displacement (cu.in.)",  
  ylab = "Miles Per Gallon (MPG)",  
  pch = 19,  
  col = "steelblue"
```

```
)
abline(
  lm(mpg ~ disp, data = mtcars),
  col = "firebrick",
  lwd = 2
)
```

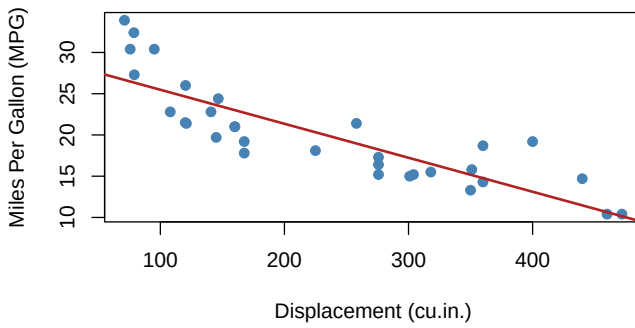


Figure 1.1: Displacement vs. Miles Per Gallon (MPG)

As demonstrated in Figure 1.1 and Chapter 1, vehicles with larger engine displacement yield lower fuel economy.

1.2 Professional Styling Features

1.2.1 Callout Boxes

Technical books use callouts to highlight important reader notes:

i Important Note for Readers

The `mtcars` dataset was extracted from the 1974 *Motor Trend* US magazine, comprising fuel consumption and 10 aspects of automobile design.

! Common Pitfall

Never evaluate models without checking for missing values (NA) or extreme outliers first!

1.2.2 Hiding Code but Showing Results

Publishers often want clean graphs without printing the underlying script in the main text:

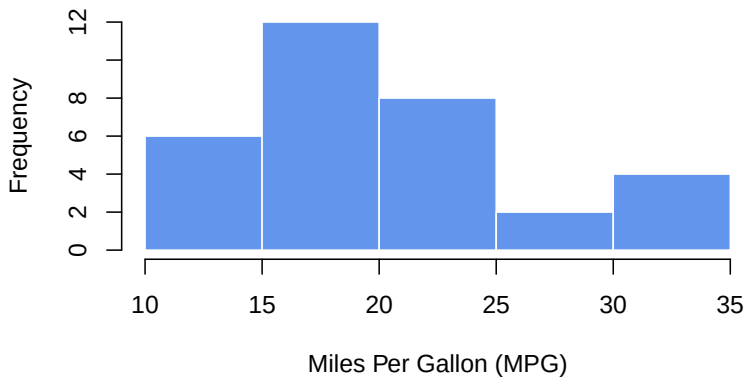


Figure 1.2: Fuel Efficiency Distribution (Code Hidden)

2 Summary & Financial Case Study

This chapter illustrates loading external CSV datasets and formatting publication-ready accounting tables.

2.1 Quarterly Performance Analysis

We load company financials from an external CSV file stored in the project data directory.

```
# Read the external CSV file
fin_data <- read.csv("data/financials.csv")

# Compute Net Profit dynamically
fin_data$Net_Profit <- fin_data$Revenue -
  fin_data$Expenses

# Display with knitr::kable
knitr::kable(fin_data)
```

Table 2.1: Quarterly Revenue vs Expenses (USD)

Quarter	Revenue	Expenses	Net_Profit
Q1	45000	31000	14000
Q2	52000	34000	18000
Q3	61000	39000	22000
Q4	78000	42000	36000

As highlighted in Table 2.1, profitability expanded across all four operating quarters.

References